

Exploratory Data Analysis (1)

July 8 and 9, 2026

For academic use exclusively within the UC Davis COSMOS

Introduction

- Exploratory Data Analysis (EDA) is the process of investigating datasets to **summarize their main characteristics** — typically through **visualization** and **statistics** — before formal modeling or inference.
- It's like being a detective: you don't jump to conclusions; instead, you explore patterns, spot anomalies, and generate hypotheses.
- Understand your dataset's structure and contents
- Detect **missing values**, **outliers**, and **errors**
- Identify **patterns**, **trends**, and **relationships**
- Make decisions about **data cleaning** and **feature selection**
- Choose the right modeling techniques

Task

Understand structure

Summary statistics

Univariate distributions

Categorical analysis

Bivariate relationships

Correlation analysis

Group-wise summaries/multivariate analysis

Dimensionality reduction

Clustering and segmentation

Clustering visualization

Hypothesis generation

Tools & Techniques

.info(), .shape, .columns, .dtypes, .head()

.describe(), mean, median, std, IQR, min/max

Histograms, boxplots, density plots

Value counts, bar plots, pie charts

Scatter plots, grouped boxplots, violin plots, faceting

.corr(), heatmaps, pairplots

groupby, aggregation by category

PCA, t-SNE, DNN (for visualization and structure discovery)

KMeans, DBSCAN, hierarchical clustering, Gaussian Mixture Models (GMM)

PCA/t-SNE scatter plots, cluster heatmaps, dendrograms, parallel coordinates

Observations and questions based on patterns or subgroup differences

Task 1: Understand Structure

Get a quick overview of the dataset's structure:

- Dimensions (rows × columns)
- Column names and types
- Glimpse at the data values

Feature

Number of rows

Number of columns

Data types

Column names

Glimpse of values

Why it matters

Affects computational choices

Helps judge complexity

Guides visualization and modeling

Watch for typos, ambiguous or duplicated names

Spot strange formats, early signs of dirty data

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Example: real data and simulated data

- Dataset one: Titanic - Machine Learning from Disaster
- <https://www.kaggle.com/c/titanic/data>
- Dataset two: simulated dataset
- Customer behavior on some product

Figure out:

- How many rows and columns are there?
- Which variables are numerical? Which are categorical?
- Are there columns you'd ignore (e.g., ID, ticket)?
- Do any values look unusual or unexpected



Task 2: Summary Statistics

Use descriptive statistics to **summarize each variable** in the dataset.

This provides insight into:

- Central tendency (mean, median)
- Spread (standard deviation, IQR)
- Min/max and range
- Possible skewness or outliers (visually or numerically)

Statistic

Mean/Median

Min/Max

Std/IQR

Counts

Unique Values

Interpretation Example

Is the variable skewed? Mean >> median?

Are there suspiciously high/low values?

How spread out is the data?

Are some categories rare? Balanced?

Detect ID columns or categorical variables

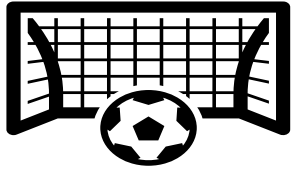
Why they are helpful?

- Understand Central Tendency
- Measure Spread / Dispersion
- Identify Outliers

Feature	Mean	Std Dev	Min	Max
Price (\$)	450K	120K	150K	950K
Sqft	1900	550	800	5000
Bedrooms	3.2	0.9	1	6

- Houses vary a lot in price and size (→ diverse market).
- Bedrooms are less variable (most homes are 3BR).
- Price per sqft can be computed to identify high-value neighborhoods.

Example



Age, fare, sex, pclass, embarked

Column

What to Look For

Possible Insight

age

Is median lower than mean? Are many children onboard?

Skewed age distribution; prioritize young passengers in rescue?

fare

High std? Large difference between min and max?

Wide fare range → socioeconomic diversity

sex

More males or females?

Gender distribution might relate to survival rate

pclass

Which class had most passengers?

Most were in 3rd class → may link to survival/access differences

embarked

Are most passengers from one location?

Could suggest boarding preference/location demographics

Task 3: Univariate Distributions

A **univariate distribution** describes how the values of a **single variable** are distributed across the dataset.

1. The Typical Values: Are most observations centered around a mean or median?

- Example: Most Titanic passengers were in their 20s–40s.

2. Spread and Variability: Is the variable tightly concentrated or widely spread? This affects model performance, normalization choices, and more.

- Example: If income varies from \$10K to \$200K, it's highly spread and may need transformation.

3. Skewness and Shape Is the distribution symmetric or skewed (long tail on one side)?

- Example: fare on the Titanic is **right-skewed** — most people paid little, but a few paid a lot.
- Skewed variables may need **log or power transformations**.

4. Presence of Outliers: Outliers may reflect data entry issues or interesting rare cases.

- Visual tools like **boxplots** help identify these.

5. Modality: Does the distribution have one peak (unimodal) or several (bimodal/multimodal)?

- Multimodal distributions suggest **subgroups** in the data (e.g., male/female heights).

6. Categorical Balance: For categorical variables, univariate analysis reveals **class balance**.

- Useful for classification problems (e.g., is purchased 90% zeros and only 10% ones?)

Variable Type

Plot Type

Description

Numeric

Histogram

Shows frequency distribution; use bins to control resolution

Numeric

KDE plot

Smooth estimate of the distribution; useful for seeing shape

Numeric

Boxplot

Shows median, IQR, and potential outliers

Numeric

Violin plot

Combines boxplot and KDE; great for comparing groups

Numeric

Rug plot

Small ticks to show actual data points (usually combined with KDE)

Numeric

ECDF plot

Empirical cumulative distribution function — shows quantile behavior

Categorical

Bar chart

Frequency of each category (count-based)

Categorical

Pie chart (limited)

Proportional view, best with few categories

Categorical

Waffle plot

Grid-style proportion visualization (requires pywaffle or Plotly)

Categorical

Treemap

Area-based representation for many categories (squarify or Plotly)



- A **histogram** shows the **frequency** (count) of data points falling within specified ranges (“bins”). It's a bar chart for continuous/numeric variables.
- Shape of the distribution, Skewness (left/right), Gaps, spikes, and multiple peaks (modality), Outliers (if bins go far into tails)
- Choosing bin width is important (too few = oversimplified; too many = noisy)
- A **KDE plot** estimates the **probability density function (PDF)** of a continuous variable. Think of it as a smooth version of a histogram — it shows where values are concentrated without binning.
- A **boxplot** summarizes the **distribution of a numeric variable** using five key statistics: Minimum, 1st quartile (Q1), Median (Q2), 3rd quartile (Q3), Maximum (excluding outliers)



- A **violin plot** is a hybrid of a **boxplot** and a **KDE plot**: **detailed distribution info + summary**
- An **ECDF plot** shows the proportion of observations **less than or equal to** each data point. It's a cumulative version of a histogram or KDE: **percentile thresholds** (e.g., what value corresponds to the 90th percentile?); compare **two or more distributions**
- A **bar chart** displays the **frequency (or proportion)** of each category in a categorical variable: Class imbalance in classification tasks
- A **treemap** is a space-filling visualization where each category is represented as a **rectangle**, and its area is proportional to the **frequency or size** of that category.

Task 4 Categorical Analysis

- Categorical analysis is the process of exploring and summarizing variables that represent **discrete groups or labels**, rather than numeric values. These variables tell you *what type or class* an observation belongs to.

Insight

Class balance

Dominant groups

Group sizes

Relationships to other variables

Feature usefulness

Example from Titanic

How many passengers are male vs. female

Most passengers embarked from Southampton

`value_counts()` shows how common each is

Survival rate by class, sex, or embarked

A column with 1 unique value is not useful

Task

Count categories

Proportions

Number of unique categories

List categories

Bar chart

Pie chart

Group comparison

- Frequency Counts
- Proportions
- Number of Unique Categories
- Unique Values
- Grouped Summaries

Tools / Code Example

```
df['var'].value_counts()
```

```
df['var'].value_counts(normalize=True)
```

```
df['var'].nunique()
```

```
df['var'].unique()
```

```
sns.countplot(x='var', data=df)
```

```
df['var'].value_counts().plot.pie()
```

```
df.groupby('category')['other_var'].mean()
```

Example

		Survival Rate by Sex	Survival Rate by Class
1	female	1.0	
2	male	0.2	
3	First		1.0
4	Second		1.0
5	Third		0.3333333333333333 3

